

M002 Week 1 — English Worksheet (Beginner)

Topic: While Loops — Move Until You Hit a Wall · **Course:** Maze Madness · **Level:** Beginner · **Time:** about 30 minutes

This week your agent uses a **while loop**: it keeps moving forward **until** it hits a wall.

1 · Predict 🧠

Read the steps. Before you imagine the agent doing it, circle or write your answer.

```
while no wall ahead:
    move forward
```

When does the agent stop? Circle one: after 3 steps · when it reaches a wall · never

Why?


```
move forward
move forward
move forward
```

The wall is only 2 steps away. What happens on the third step?

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The agent should keep moving **until** it hits a wall.

```
# clean
while no wall ahead:
    move forward
```

```
# buggy
while wall ahead:
    move forward
```

What is wrong? When does the buggy agent move?

Pair B — The agent should check the wall **before every step**.

```
# clean
while no wall ahead:
    move forward
```

```
# buggy
while no wall ahead:
    move forward
    move forward
```

What is wrong? What can happen right in front of the wall?

3 · Fill the Gap

The agent should walk forward until it reaches the wall. One word is missing. Fill it in using the word bank.

```
while ____ wall ahead:  
    move forward
```

Word bank: no · big · one

Write the missing word:

4 · Tell Me What You Built 📷

Now switch to your homework world. Use a **while loop** to move the agent down the straight maze until it reaches the wall. When you finish, come back here.

Send a photo or video of the agent at the wall, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My while loop checked ...

The agent stopped because ...

A while loop is useful because ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while the agent runs the maze. Talk like you are teaching a friend. Try to use these words: **while**, **loop**, **wall**, **forward**.

1. Show the start of the maze, then run your code.
2. Read your while loop out loud and say what makes it stop.
3. Say why a while loop is better than writing `move forward` many times.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.